



ALLEN
CAREER INSTITUTE
KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
28-02-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

PAPER-2

Time : 3 Hours

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

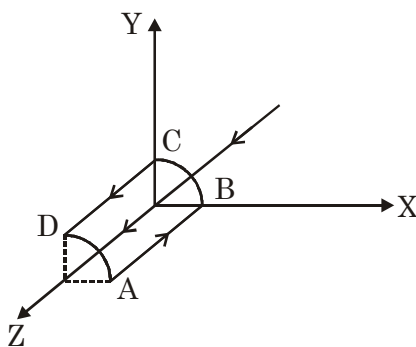
HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 24)

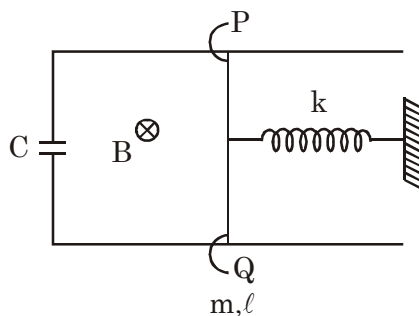
- This section contains **SIX** questions.
 - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
 - For each question, choose the correct option(s) to answer the question.
 - Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
 - **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.
1. A long thin wire is carrying current along positive Z-axis. Another current carrying loop made of uniform wire is arranged with its vertices at A(a, 0, a), B(a, 0, 0), C(0, a, 0), D(0, a, a) as shown in figure. BC and DA are in the shape of a quarter circle centered on Z-axis. The loop is constrained to rotate about Z-axis without friction.



- (A) Mutual inductance between the long wire and the loop is zero.
 (B) Net magnetic force acting on the loop due to the long wire is zero.
 (C) Net torque due to magnetic force on the loop about Z-axis is zero.
 (D) Consider an ampere loop in the shape of a circle of radius 'a'. Plane of the loop is parallel to XZ-plane with its centre at (0, a/2, 0). Let $\vec{B}$ be the net magnetic field in space. Then $\oint \vec{B} \cdot d\vec{\ell}$ around the Ampere loop is zero.

Space for Rough Work

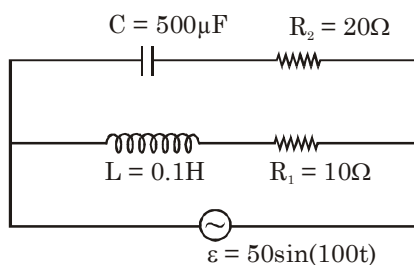
2. A charged particle with velocity $\vec{v} = x\hat{i} + y\hat{j}$ moves in a magnetic field $\vec{B} = y\hat{i} + x\hat{j}$. The magnitude of force acting on the particle is F . Which one of the following statement are correct ?
- (A) No force will act on particle if $x = y$.
 (B) $F \propto (x^2 - y^2)$ if $x > y$.
 (C) The force will act along z -axis if $x > y$.
 (D) The force will act along y -axis if $y > x$.
3. An uncharged capacitor of capacitance 'C' is connected to smooth parallel horizontal perfectly conducting rails. A perfectly conducting rod PQ of mass m and length ℓ is connected to one end of a massless, non-deformed spring of stiffness K whose other end is fixed. A uniform vertical magnetic field B is existing in the region as shown. Now the rod is initially displaced towards right by d and then released from rest. Then choose the correct option(s).



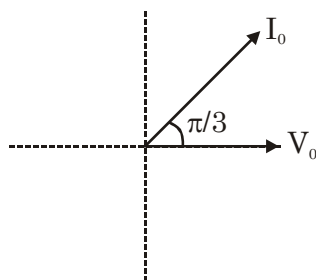
- (A) The time period of SHM executed by the rod is $2\pi\sqrt{\frac{m + CB^2\ell^2}{K}}$.
- (B) The maximum velocity of rod is $d\sqrt{\frac{K}{m + CB^2\ell^2}}$.
- (C) The maximum velocity of rod is $d\sqrt{\frac{K}{m}}$.
- (D) The maximum charge on the capacitor is $CB\ell d\sqrt{\frac{K}{m + CB^2\ell^2}}$.

Space for Rough Work

4. In the AC circuit shown in the figure :



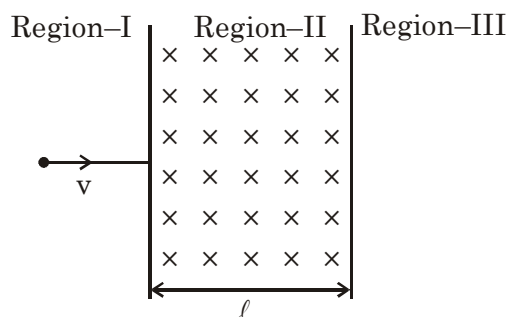
- (A) The net impedance of circuit is $4\sqrt{10}\Omega$.
- (B) The rms current through resistor R_1 is $\frac{5}{2}\text{A}$.
- (C) The rms current through resistor R_2 is $\frac{5}{4}\text{A}$.
- (D) The power developed in the resistor R_1 is 62.5 watt.
5. For an LCR series circuit, phasors of current I and applied voltage $V = V_0\sin\omega t$ are shown in diagram at $t = 0$. Which of the following is/are correct?



- (A) At $t = 0$, instantaneous power supplied by source is negative.
- (B) In the interval $0 < t < \frac{2\pi}{3\omega}$, average power supplied by source is positive.
- (C) At $t = \frac{5\pi}{6\omega}$, instantaneous power supplied by source is negative.
- (D) If ω is increased slightly, angle between the two phasors decreases.

Space for Rough Work

6. A particle of mass m and charge q , moving with velocity u enters Region-II normal to the boundary as shown in the following figure. Region-II has a uniform magnetic field B . Choose the correct choice(s).



- (A) The particle enters region-III only if its velocity $v > \frac{q\ell B}{m}$.
- (B) The particle enters region-III only if its velocity $v < \frac{q\ell B}{m}$.
- (C) Path length of the particle in region-II is maximum when velocity $v = \frac{q\ell B}{m}$.
- (D) Time spent in region-II is same for any velocity v as long as the particle returns to region-I.

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

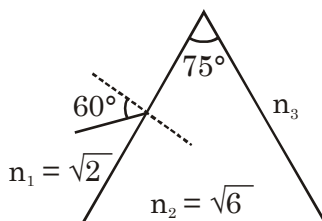
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
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9	9	9	9	9	9	9	9	9	9

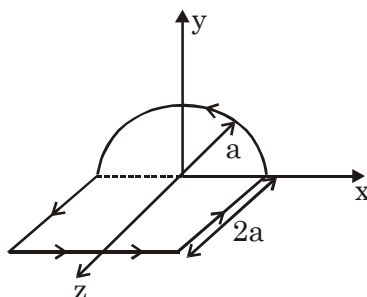
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A slab of glass of thickness 6 cm and index 1.5 is placed somewhere in between a concave mirror and a point object, perpendicular to the mirror's optical axis. The radius of curvature of the mirror is 40 cm. If the reflected final image coincides with the object, find the distance (in cm) of the object from the mirror.
2. A prism is kept in two different media as shown in figure. The prism is made of material of refractive index $\sqrt{6}$. The ray undergoes a deviation of 45° , then n_3 is :



Space for Rough Work

3. On a planet a body trajectory is given by $y = x - 0.8x^2$, here x and y are horizontal and vertical positions in meters. If the body is thrown vertically upwards with same speed of projection, the distance travelled (in m) by the body during its flight is (Neglect any frictional force on body and gravity is uniform over the planet)
4. If the current in the loop in the figure shown below is 1 Amp, what is the magnitude of the magnetic field at the origin? Take $a = 1\text{m}$. If the magnetic field is B (in T), fill $B \times 10^7$ in OMR sheet.



5. In a cyclotron α -particles (which are doubly ionized helium atoms) are accelerated. The radius of the greatest circular path is 50 cm. The magnitude of magnetic flux density is 1 Tesla. To what greatest energy (in MeV) can these particles be accelerated in this cyclotron? Take mass of helium = $6.4 \times 10^{-27}\text{ kg}$.

Space for Rough Work

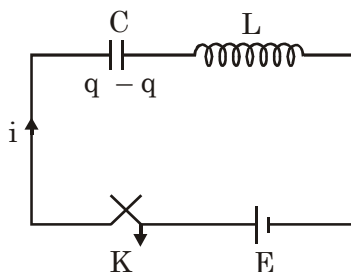
6. A resistor of constant resistance is connected to the generator of a bicycle (sometimes wrongly named as "dynamo"). To rotate the generator at a constant rotational speed of revolutions n and $2n$, the average torque to be applied is M and $1.8M$, respectively. What is the average torque (in multiples of M) in order to rotate the generator at a rotational speed of $3n$? The mechanical losses of the generator can be neglected.
-

Space for Rough Work

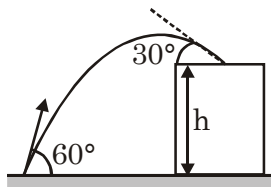
SECTION-III : (Maximum Marks : 18)

- This section contains **SIX** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. In the figure shown below, the initial charge on the capacitor is $q = -CE$ and there is no initial current. What is the work done by the battery till the current in the inductor first becomes maximum? If the work done by the battery is $n \times \frac{1}{2} CE^2$, Fill n in OMR sheet.

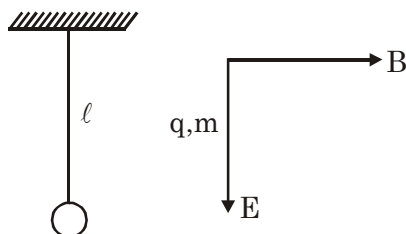


2. A stone projected at an angle of 60° from the ground level strikes at an angle of 30° on the roof of a building of height 'h'. If the speed of projection of the stone is $\sqrt{Ngh}$, find N.



Space for Rough Work

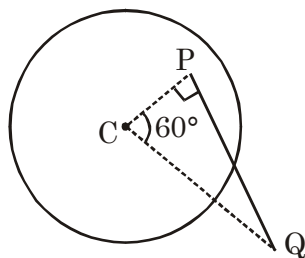
3. In the region having mutually perpendicular electric and magnetic fields, a dielectric ball is attached to a thread of length ℓ (see figure). Mass of the ball is m , charge q . The electric field E , magnetic field B . Find the approx. period (in sec) of small oscillations of the pendulum in a plane perpendicular to the magnetic field (Take $\ell = 1\text{m}$, $q = -1\mu\text{C}$, $E = 7.5\text{ V/m}$, $m = 1\text{mg}$ and $B = 0.5\text{ T}$)



4. A magnetic field, confined in a cylindrical region of radius R is changing at the rate of 4 T/s .

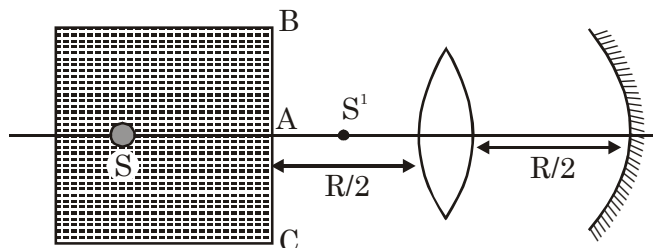
A conducting rod PQ of length $\sqrt{\frac{3}{2}}R$ is placed in the region as shown. The two ends of the rod are now joined by straight conducting wires to the center C , so that the circuit is completed. The resistance of this entire loop is r . At time t , the magnetic field in the cylindrical region

is B . At time t , magnetic force acting on the rod is given by $\frac{BR^3}{ar}(\pi + b)\sqrt{2}$. Find $(a - b)$.



Space for Rough Work

5. A thin walled glass container is filled with water ($\mu = \frac{4}{3}$). A point source S is fixed inside water. A thin equiconvex lens ($\mu_l = 1.5$) and concave mirror are arranged with mutual separation $R/2$ as shown in figure. If the distance of the source from the point A is $\frac{2R}{3}$, the separation between the source and final image as seen by an observer at the position of S is $nR/6$, find n. **All curved surfaces of lens and mirror have radius of curvature R.**



6. The speed of a motorboat in still water is four times the speed of a river. Normally, the motorboat takes one minute to cross the river to the port straight across on the other bank. One time due to a motor problem, it was not able to run at full power, and it took four minutes to cross the river along the same path. Relative speed of the motor boat in the situation of motor problem is $\frac{\sqrt{31}}{x}$ times of its usual relative speed. Find the value of X. (Assume that the speed of the water is uniform throughout the whole width of the river)

Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I : (Maximum Marks: 24)

- This section contains **SIX** questions.
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1. Photons of wavelength 12.4 nm were allowed to strike a metal plate having work function 25 eV. Identify correct statement(s) :
 (A) For photoelectrons : $KE_{\max} = 75 \text{ eV}$
 (B) Minimum wavelength of emitted photoelectrons is nearly $1.41 \text{ \AA}$
 (C) For all photoelectrons emitted : $KE = 75 \text{ eV}$
 (D) For all emitted photoelectrons : wavelength = $1.41 \text{ \AA}$
2. Which of the following ligand is/are ambidentate ligand?
 (A) $\text{SCN}^\ominus$ (B) NO_2^- (C) $\text{S}_2\text{O}_3^{2-}$ (D) $\text{C}_2\text{S}_2\text{O}_2^{2-}$
3. Which of the following is/are correct order of nucleophilicity in aqueous medium?
 (A) $\text{CH}_3^\ominus > \text{NH}_2^\oplus > \text{OH}^\ominus > \text{F}^\ominus$ (B) $\text{F}^\ominus > \text{Cl}^\ominus > \text{Br}^\ominus > \text{I}^\ominus$
 (C) $\text{OH}^\ominus < \text{SH}^\ominus$ (D) $\text{EtO}^\ominus > \text{CH}_3\text{COO}^\ominus$

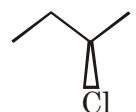
Space for Rough Work

4. Select the correct option for 1st order reaction $2A \rightarrow \text{product}$ having rate constant of reaction $1.386 \times 10^{-2} \text{ min}^{-1}$.

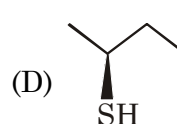
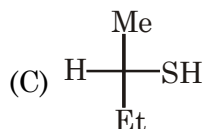
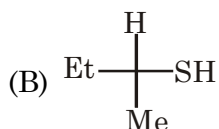
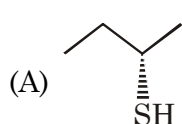
(A) time required for 75% completion of reaction is 50 min.
 (B) time required for 99.9% completion of reaction is approximately 250 min.
 (C) Rate of disappearance of 'A' at concentration 0.1 M is $2.772 \times 10^{-3} \text{ M/min}$.
 (D) Rate of reaction at $[A] = 0.1 \text{ M}$ is $1.386 \times 10^{-3} \text{ M/min}$.

5. Which of the following compound undergo hydrolysis to give hydracid?

(A) POCl_3 (B) P_4O_{10} (C) BiCl_3 (D) NCl_3

6.  $\xrightarrow{\text{NaSH}}$ product :

major product of the reaction is/are :



Space for Rough Work

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+									
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3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	●	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	●	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
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9	9	9	9	9	9	9	9	9	9

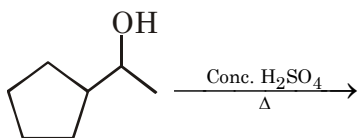
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Unit cell $a = b \neq c$ and $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$, $a = 4\sqrt{6}\text{\AA}$ then distance in $\text{\AA}$ of nearest void from center of particle that form lattice ?
- Consider the following molecules.
 SO_2Cl_2 , XeF_2 , XeF_4 , BrF_5 , SO_3 , POCl_3 , Cr_2S_3 , AsCl_3 , SF_4 , SeF_6 , ICl , CrO_2Cl_2 , N_2O_4 , SiF_4
 (i) if the total number of molecules which are hydrolysed by redox reaction = 'X'
 (ii) if the total number of molecules which produce atleast one diprotic oxyacid on complete hydrolysis = 'Y'
 (iii) if the total number of molecules which produce atleast one non acidic product on hydrolysis = Z

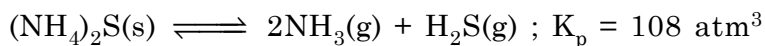
then find the value of $\frac{Y+Z}{X} = ?$

Space for Rough Work

3. Identify number of α -hydrogen present in final major product in given reaction



4. For the equilibrium of the reaction :



Solid $(\text{NH}_4)_2\text{S}$ is in equilibrium with its product and equilibrium gets disturbed due to addition of H_2S . If the pressure of H_2S at new equilibrium become 4 times the old equilibrium than what is ratio of new total pressure to old total pressure at equilibrium.

5. Consider the following statements.

- (i) Electron population in π^* orbital of free CO is lower than in its complex $[\text{Cr}(\text{CO})_6]$.
- (ii) $\text{Co}(\text{CO})_4$ attains stability by oxidation of Co and dimerization
- (iii) In $[\text{Ti}(\text{CO})_6]^{-2}$ C–O bond energy is greater than C–O bond energy in $[\text{Mn}(\text{CO})_6]^{\oplus}$
- (iv) In $[\text{V}(\text{CO})_6]$ M–C bond length is greater than in M–C bond length in $[\text{V}(\text{CO})_6]^{-1}$
- (v) C–O bond order in $[\text{Cr}(\text{CO})_6]$ is lower than CO
- (vi) CO, C_2H_4 , $\text{NO}^{\oplus}$, PR_3 are ' σ ' donor and ' π '-acceptor ligands

if correct statements are 'x' and incorrect statements are 'y' than find the value of $\frac{x}{y} = ?$

6. 2-pentene $\xrightarrow{\text{Br}_2 / \text{CCl}_4}$ product mixture

Identify total number of dibromo products in product mixture?

Space for Rough Work

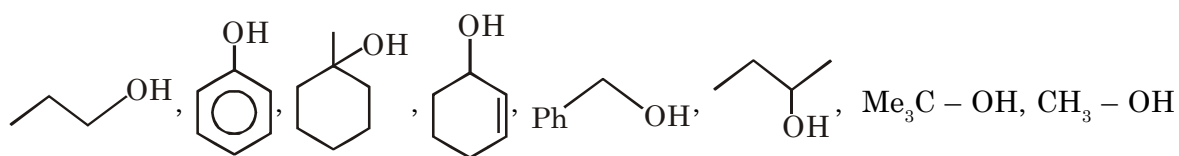
SECTION-III : (Maximum Marks : 18)

- This section contains **SIX** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

-
1. For reaction (i) $A \rightarrow B$ and (ii) $P \rightarrow Q$, if k_1 & k_2 are rate constant and E_{a1} , E_{a2} are activation energy of reaction (i) & (ii) respectively, find $(E_{a2} - E_{a1})$ (in kcal/mole) at 227°C .
 $k_1 = 100 k_2$
 $10A_1 = A_2$
 $\ln 10 = \frac{7}{3}$
2. Find out the number of following orders which are **CORRECT** against the mentioned properties.
- (i) $\text{I}_2 > \text{Br}_2 > \text{Cl}_2 > \text{F}_2$: oxidising strength
 - (ii) $\text{As}^{+3} > \text{Bi}^{+3}$: reducing strength
 - (iii) $\text{HgF}_2 > \text{HgCl}_2 > \text{HgBr}_2 > \text{HgI}_2$: solubility in water
 - (iv) $\text{NaI}_3 > \text{KI}_3$: thermal stability
 - (v) $\text{NaF} < \text{MgF}_2 < \text{AlF}_3$: lattice energy
 - (vi) $\text{KF} > \text{RbF} > \text{CsF} > \text{AgF}$: Covalent character
-

Space for Rough Work

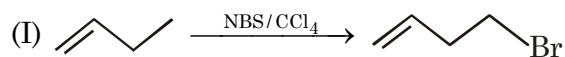
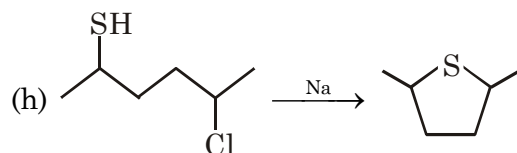
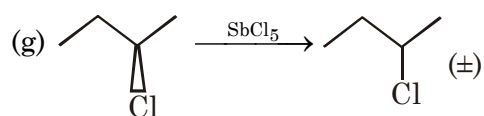
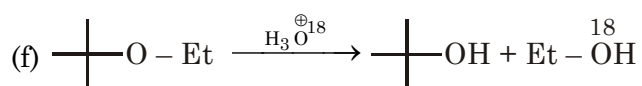
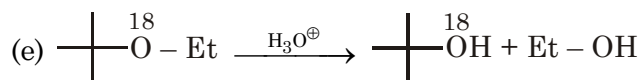
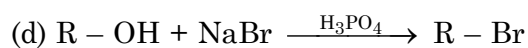
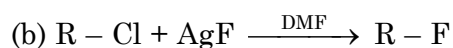
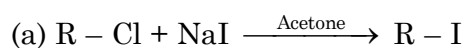
3. Identify total number of compounds which give immediate turbidity on reaction with HCl/ZnCl_2 ?



4. In an aqueous solution mole fraction of solute is 0.2 and its density is d_1 (g/ml). It is diluted such that mole fraction of solute becomes 0.1 and density becomes d_2 g/ml. If density of new solution is 20% less than initial solution then molarity of new solution is almost half of the original solution. If molar mass of the solute is 'X' gm/mol, then closest integer value of $\left[\frac{X}{13} \right]$ is
5. Find the total number of complexes are following EAN rule. (Atomic number : $\text{Mo} \Rightarrow 42$, $\text{Pt} \Rightarrow 78$, $\text{Ag} = 47$)
- $[\text{Mo}(\sigma\text{-C}_3\text{H}_5)\text{Br}(\text{NH}_3)_2]$, $[\text{Fe}(\eta^5\text{-C}_5\text{H}_5)_2]$, $[\text{Ni}(\text{DMG})_2]$, $[\text{Mn}_2(\text{CO})_{10}]$, $[\text{Fe}(\text{CO})_2(\text{NO})_2]^0$,
 $[\text{PtCl}_3(\eta^2\text{-C}_2\text{H}_4)]^0$, $[\text{Ag}(\text{CN})_2]^{-1}$, $[\text{Fe}_2(\text{CO})_9]^0$

Space for Rough Work

6. Number of reactions which are correctly matched with their major product?



Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I : (Maximum Marks: 24)**

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

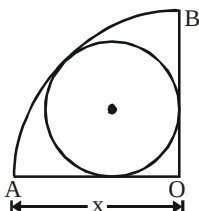
1. Sector OAB is a quarter of circle of radius x . A circle of radius ' r ' is drawn inside this sector tangent to three points as shown in figure, then which of the following is/are true ? (where r & x are variable)

(A) $\left(\frac{dr}{dx}\right)^2 + \left(\frac{dx}{dr}\right)^2 + 2\frac{d^2x}{dr^2} = 6$

(B) $\left(\frac{dr}{dx}\right)^2 + \left(\frac{dx}{dr}\right)^2 - 2\frac{d^2x}{dr^2} = 8$

(C) $\frac{x+r}{x-r} + \frac{r-x}{x+r} = 2$

(D) $\sum_{n=1}^{10} \frac{d^n r}{dx^n} = \frac{r}{x}$



Space for Rough Work

2. Let $f(x) = \begin{cases} -2 & \text{for } -3 \leq x \leq 0 \\ x-2 & \text{for } 0 < x \leq 3 \end{cases}$, $g(x) = [f(|x|)] + [|f(x)|]$ where $[.]$ denotes the greatest integer function. Which of the following statement(s) is/are true ?
- (A) $g(1.7) = -1$
 (B) $g(x)$ is not differentiable at any point
 (C) $g(x)$ is continuous at all non integral points
 (D) $g(x)$ is differentiable at some points
3. If $f(x) = \begin{cases} 2x^2 - 3x + 2a & ; x \in \mathbb{Q} \\ x^2 - bx & ; x \notin \mathbb{Q} \end{cases}$ is continuous at $x = 1$ and $x = 3$, then (where $\mathbb{Q}$ is the set of rational numbers)
- (A) $a = -1$ (B) $a = \frac{3}{2}$ (C) $b = -1$ (D) $b = \frac{3}{2}$
4. If $\frac{\cos^4 x}{\theta_1} + \frac{\sin^4 x}{\theta_2} = \frac{1}{(\theta_1 + \theta_2)}$, then $\frac{d\theta_2}{d\theta_1}$ equals (where x is not dependent on θ_1 or θ_2)
- (A) $\sin^2 x$ (B) $\cos^2 x$ (C) $\tan^2 x$ (D) $\sec^2 x$

Space for Rough Work

5. A right angled triangle has one leg of length 1, another of length 'x' and hypotenuse of length 'y'. The angle opposite to the side x is θ then -

(A) $y^n - x^n = \frac{1 - \sin^n \theta}{\cos^n \theta}$

(B) $\lim_{\theta \rightarrow \frac{\pi}{2}} (y - x) = 0$

(C) $\lim_{\theta \rightarrow \frac{\pi}{2}} (y^2 - x^2) = 1$

(D) $\lim_{\theta \rightarrow \frac{\pi}{2}} (y^3 - x^3)$ does not exist.

6. Value of $\lim_{x \rightarrow 1} \frac{(\log(1+x) - \log 2)(3 \cdot 4^{x-1} - 3x)}{((7+x)^{1/3} - (1+3x)^{1/2}) \sin \pi x}$ is equal to

(A) $\frac{9}{\pi}(2\log 2 - 1)$

(B) $\frac{9}{4\pi}[\log 4 - 1]$

(C) $\frac{9}{2\pi} \left[\log 4 - \frac{1}{2} \right]$

(D) $\frac{2}{3\pi}[2\log 2 - 1]$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- $\lim_{x \rightarrow -\infty} \left(\sqrt{x^2 - \frac{4x}{3}} + x + 2 \right)$ is equal to-
- Value of $\lim_{x \rightarrow \frac{3\pi}{4}} \frac{4\sin^2 x \cos x - \cos x + \sin x}{\sin x + \cos x}$ is
- If $f(x) = \begin{cases} \frac{\sin^{-1}(\{x\}^2 + \{x\}^3) - x}{\{x\}}, & x \neq 0 \\ -1, & x = 0 \end{cases}$, then the value of RHD of $f(x)$ at $x = 0$, is
 (where $\{.\}$ denotes fractional part function)

Space for Rough Work

4. If $\sum_{n=0}^{\infty} \arccot \left(\frac{n^2 + n + 4}{2} \right) = k\pi$, then value of k is equal to
5. Let the function $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = \min.(x + 2, 4 - 2x, 1 + 4x)$. The maximum value of $f(x)$ is equal to -
6. A function $f : A \rightarrow B$ is defined such that Set A consists of 6 different elements and set B consists of 4 different elements, then number of onto functions f is
-

Space for Rough Work

SECTION-III : (Maximum Marks : 18)

- This section contains **SIX** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. If a derivable function $y = f(x)$ satisfies the condition $f(x) + f(y) = f(x)f(y) + f(xy) \forall x, y \in \mathbb{R}$ where $f(1) = 0$ and $f'(1) = -2$, then the number of solution(s) of the equation $f(x) = 2^x$ is
2. Let $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4$ where $a_i \in \mathbb{R}$ and $\lim_{x \rightarrow 0} \left(\frac{f(x) - x + 2x^2}{3x^3} \right)^{\frac{2}{x}} = e^4$, then $\sum_{i=0}^4 a_i$ is equal to
3. Let $f(x) = x + 1$, $g^{-1}(x) = x^3 + x + 1$ and $h(x) = g(f(x))$, then $\lim_{x \rightarrow 1} \frac{h^{-1}(x) - h^{-1}(1)}{x - 1}$ is equal to

Space for Rough Work

-
4. Let $P(x) = x^{10} + ax^8 + bx^6 + cx^4 + dx^2 - 1$ be polynomial with real coefficients. If $P(1) = 1$, $P(2) = -1$ then the minimum number of real zeroes of $P(x)$ is
5. Let $f(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5}$ and $g(x) = f^{-1}(x)$, then $|g''(0)|$ is
6. Number of values of x in $[0, \pi]$ for which $\sin^{-1}(\sin 6x) = \cos^{-1}(\cos x)$ is -
-

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has three sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	24
II	Numerical Value Type (Up to second decimal place)	6	+4 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24
III	Single digit Integer (0-9)	6	+3 If only the bubble corresponding to correct answer is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	18

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator